

Open-ladder identities

Cross-readings from the frozen field definitions

Abstract

Algebraic consequences of the frozen definitions $B_{ij} = \Re(\psi_i^* \psi_j)$, $H = -A$, the sum map, and the earned dispersions of J_2 and J_3 . The same identity often sits in more than one part of the ledger. Those overlaps are collected in the first part. Later parts record the local derivation for each open hypothesis. Nothing here is a campaign.

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Part I

Cross-readings

The ledger splits cleanly into campaigns. Several of the open rows are the same formula read on different objects. This part records those coincidences. A local derivation is repeated, in shorter form, under the hypothesis it came from.

1 One bond number

1.1 The identities

Let $\psi_i, \psi_j \in \mathbb{C}$ and write

$$B_{ij} = \Re(\psi_i^* \psi_j), \quad q_{ij} = |\psi_i|^2 + |\psi_j|^2, \quad s = \psi_i + \psi_j, \quad d = \psi_i - \psi_j. \quad (1)$$

The sum-map convention in `split0.fiber_point` is the same s and d , up to which daughter is called i . The swap $d \mapsto -d$ is the endpoint gauge already quotiented by STORE.

Lemma 1.

$$\begin{aligned} |s|^2 &= q_{ij} + 2B_{ij}, & |d|^2 &= q_{ij} - 2B_{ij}, \\ |s|^2 - |d|^2 &= 4B_{ij}, & |s|^2 + |d|^2 &= 2q_{ij}. \end{aligned}$$

If $q_{ij} > 0$,

$$P_- = \frac{|d|^2}{|s|^2 + |d|^2} = \frac{1}{2} - \frac{B_{ij}}{q_{ij}}, \quad P_+ = \frac{1}{2} + \frac{B_{ij}}{q_{ij}}. \quad (2)$$

Furthermore $|B_{ij}| \leq q_{ij}/2$, with equality if and only if ψ_i and ψ_j are nonnegative real multiples of one complex phase.

Proof. Expand $|\psi_i \pm \psi_j|^2$. The cross term is $2\Re(\psi_i^* \psi_j)$. Cauchy–Schwarz gives the bound on B_{ij} . The unitary modes $\alpha = s/\sqrt{2}$ and $\beta = d/\sqrt{2}$ produce the same ratio P_- , because the factors of 2 cancel. $\square$

Corollary 2. For $q_{ij} > 0$,

$$B_{ij} = 0 \iff |s| = |d| \iff P_+ = P_- = \frac{1}{2}.$$

The sum-map norm change is $\Delta\|\psi\|^2 = 2B_{ij}$ (*accounting.event.ledger*). Hence $B_{ij} = 0$ if and only if that contraction is norm-neutral. The energy change is $\Delta E_\psi = 2B_{ij} - 2\Sigma_{\text{cross}}$. On K_N every third vertex is a common neighbour, so $\Sigma_{\text{cross}} = 0$ and norm neutrality coincides with energy neutrality. On a cut leg the exclusive-neighbour sum survives, and the two neutralities diverge.

If $q_{ij} = 0$ both amplitudes vanish. There is then no pair, and P_- is undefined. The limit of $h_2(P_-)$ along $q \rightarrow 0$ depends on the direction of approach, so an empty pair is not assigned the value 1 by continuity.

1.2 Binary entropy, as an optional reading

Remark 3. INFO-0 recorded counts with binary logarithms and did not introduce $-\sum p \log p$. The map $B/q \mapsto P_-$ does not need a further identification. You could still attach to an occupied pair the Shannon entropy of its $\pm$ weight,

$$s_e = h_2(P_-), \quad h_2(p) = -p \log_2 p - (1-p) \log_2 (1-p). \quad (3)$$

Then $0 \leq s_e \leq 1$, with $s_e = 1$ if and only if $B_{ij} = 0$, and $s_e = 0$ at the Cauchy–Schwarz ends. This s_e is the uncertainty of one projective $\pm$ measurement. It is not the von Neumann entropy of a pure global state. It does not see the phase of d relative to s . At $B_{ij} = 0$ that phase is a pure current: $|\Im(\psi_i^* \psi_j)| = |\psi_i| |\psi_j|$. The continuous fibre dimension $\dim_{\mathbb{R}} d = 2$ counts magnitude and phase. The number s_e counts neither as a dimension. The two are not proportional, which is the overlap between the post-STORE entropy row and the BH-Q dimension count (Section 2).

Lemma 4. With $x = B/q$ and $|x| \leq 1/2$,

$$h_2\left(\frac{1}{2} - x\right) = 1 - \frac{2}{\ln 2} x^2 - \frac{4}{3 \ln 2} x^4 + O(x^6).$$

There is no odd term.

Proof. Set $f(x) = (\frac{1}{2} - x) \ln(\frac{1}{2} - x) + (\frac{1}{2} + x) \ln(\frac{1}{2} + x)$, so $h_2 = -f/\ln 2$. Then $f(0) = -\ln 2$, $f'(0) = 0$, $f''(x) = 4/(1-4x^2)$, $f''(0) = 4$, $f'''(0) = 0$, $f''''(0) = 32$. The degree-4 Taylor polynomial is $-\ln 2 + 2x^2 + \frac{4}{3}x^4$. $\square$

1.3 Three ensembles, one pair

Theorem 5. Let ψ be Haar-uniform on the unit sphere in $\mathbb{C}^N$, $N \geq 2$, and let $\{i, j\}$ be any fixed pair. Then P_- is uniform on $[0, 1]$, and

$$\mathbb{E}[h_2(P_-)] = \frac{1}{2 \ln 2}.$$

The expectation is exact at finite N , not a large- N limit. Weighting pairs by q_{ij} does not change it: $h_2(P_-)$ depends only on the fraction, and for the Dirichlet law the fraction is independent of the pair norm.

Proof. The squared moduli of a Haar vector in $\mathbb{C}^N$ are Dirichlet with every parameter equal to 1. They have the law of $(e_1, \dots, e_N)/\sum_k e_k$ with e_k i.i.d. exponential. For any two coordinates, $e_i/(e_i + e_j)$ is uniform on $[0, 1]$. The rotation $\alpha = (\psi_i + \psi_j)/\sqrt{2}$, $\beta = (\psi_i - \psi_j)/\sqrt{2}$ is a global unitary. Haar measure is invariant, so $P_- = |\beta|^2/(|\alpha|^2 + |\beta|^2)$ is uniform on $[0, 1]$. The integral $\int_0^1 -p \ln p dp = 1/4$, and the partner term in $p \mapsto 1-p$ is equal, so

$$\int_0^1 h_2(p) dp = \frac{1}{\ln 2} \cdot \frac{1}{2} = \frac{1}{2 \ln 2}.$$

$\square$

Proposition 6. *If the two moduli are equal and the relative phase is uniform, or if the pair is a real Gaussian, then P_- is Beta($\frac{1}{2}, \frac{1}{2}$) and*

$$\mathbb{E}[h_2(P_-)] = 2 - \frac{1}{\ln 2}.$$

Proof. Equal moduli and uniform phase give $P_- = (1 - \cos \phi)/2$. The density of this law is $1/(\pi \sqrt{p(1-p)})$. A real Gaussian pair has the same norm fraction, because the argument of (g_1, g_2) is uniform. One of the two symmetric terms in the average is

$$\frac{1}{\pi \ln 2} \int_0^1 \frac{-p \ln p}{\sqrt{p(1-p)}} dp = \frac{1}{\pi \ln 2} (-\partial_a B(a, b))_{a=3/2, b=1/2}.$$

Here $B(3/2, 1/2) = \pi/2$ and $\psi(3/2) - \psi(2) = 1 - 2 \ln 2$, so $-\partial_a B = (\pi/2)(2 \ln 2 - 1)$. One term equals $(2 \ln 2 - 1)/(2 \ln 2)$. Both terms give $2 - 1/\ln 2$. $\square$

Proposition 7. *The normalised all-ones vector on K_N , which is the ground state of $H = -A$, has $B_{ij}/q_{ij} = 1/2$ on every edge, hence $P_- = 0$ and $h_2 = 0$.*

A flat-magnitude random-phase vector is Proposition 6, not Theorem 5. The two means differ: $1/(2 \ln 2) \approx 0.721$ against $2 - 1/\ln 2 \approx 0.557$.

1.4 Scale separation

Proposition 8. *Fix $v > 0$ and let $\psi_i = \sigma g$ with g complex normal, $\mathbb{E}|g|^2 = 1$, and $\psi_j = v$ real. Set $\lambda = \sigma/v$. Then*

$$x = \frac{B}{q} = \frac{\lambda \Re g}{1 + \lambda^2 |g|^2}.$$

As $\lambda \rightarrow 0$ or $\lambda \rightarrow \infty$, $x \rightarrow 0$ in probability, so $h_2(1/2 - x) \rightarrow 1$ in probability. For small λ , $\mathbb{E}[(\Re g)^2] = 1/2$ and

$$1 - \mathbb{E}[h_2] = \frac{\lambda^2}{\ln 2} + O(\lambda^4).$$

You could read this as follows. A dynamical selection of quadrature is one way to reach $h_2 = 1$. A mismatch of the two amplitudes is another, and it does not use the phase. The case $\psi_j = 0$, $\psi_i \neq 0$ is exactly $B = 0$. That endpoint is the $\psi = 0$ limit, which the ledger already separates from the JOINT vacua.

1.5 Where the same number sits

I1a assigns $\ln 2$ nats, hence one bit, to a saturated leg, and the matching theorem gives $A = 4 \ln 2 \cdot k \ell_P^2$. In bits, $S = k$ and

$$\frac{S}{A} = \frac{1}{4 \ell_P^2 \ln 2}.$$

You could take the channel set to be the k cut legs and set $S_Q = \sum_e h_2(P_{-,e})$. Then $S_Q = k$ if and only if $\bar{h} = 1$, and the coefficient above is reproduced with no free area. A Haar law on those same legs would give $S_Q = k/(2 \ln 2)$ instead, short by the constant $2 \ln 2$. The interior edges of K_N are a different set: there are $\binom{N}{2}$ of them, and Theorem 5 puts about 0.721 bit on each. Edges of K_N incident to a leg-bearing vertex number about kN . The cardinality of the set is part of the definition of the sum. The algebra does not choose the set.

BR2-QUADRATURE says that structural response tracks $\cos \Delta \theta$ through B and a staggered flux tracks $\sin \Delta \theta$ through the current. It does not set $B_{ij} = 0$. The word “quadrature” in that verdict and the condition $B = 0$ share a trigonometric function and not a theorem.

2 Kernel dimension is a different count

Lemma 9. *The sum map $(p, q) \mapsto s = p + q$ has kernel the complex line $d = p - q$, real dimension 2, for every s including $s = 0$. After the $U(1)$ quotient the physical dimension drops to 1 only for the all-zero merged state. The swap $d \mapsto -d$ is discrete and does not halve the dimension.*

A tree of $n - 1$ interior merges therefore carries naive continuous dimension $2(n - 1)$, minus all-zero exceptions. That is the count $D_Q = 2N_d + D_c$ in the ledger. Lemma 9 and Remark 3 do not convert it into h_2 .

Proposition 10. *Suppose each merge contributes an independent complex coordinate, and the exterior imprint of those coordinates factors through a complex boundary space of dimension at most $b = |\partial R|$. Then*

$$D_Q \geq 2(N_Q - b).$$

For a space-filling collapse, $N_Q = n_R - 1$, so the lower bound is volume-leading. If the exterior evolution is the collapsed graph alone, Q does not enter H (the store is frozen between events), the Jacobian on the d -coordinates vanishes, and $D_Q = 2(n_R - 1)$ up to all-zero exceptions.

An area law for this kernel would need either N_Q itself of order b , or an imprint whose rank cancels the bulk and leaves a boundary layer. Neither is implied by $H = -A$ or by the sum map. You could have a volume law for D_Q and, on a different set of edges, an area law for $\sum h_2$. They count different things: interior relative modes, and leg channels. Promoting the factor 2 in $D_Q = 2N_d$ to two bits per merge is a third count, distinct from Ila's one bit per saturated leg.

3 One length in the chain weave

The chain construction places stitch endpoints at density $\rho = 4\lambda$ on the sheets and sets

$$\ell_W = \rho^{-1/2} = \frac{1}{2\sqrt{\lambda}} \tag{4}$$

in sheet cells. A transverse step costs an in-plane walk of that typical length, plus one stitch edge.

Proposition 11. *In coordinates (x, y, z) with z the sheet index, the Riemannian approximation consistent with that cost is*

$$ds^2 = dx^2 + dy^2 + \ell_W^2 dz^2. \tag{5}$$

The linear axis ratio is ℓ_W . In-plane shifts and a transpose act on (x, y) . They do not rotate z into the sheet. The quadratic symbol of J_2 is already isotropic in the sheet ($\text{Hess} = 4I$), so those moves have no quadratic in-plane anisotropy left to erase.

Volume growth in the window $\ell_W \ll r \ll S\ell_W$ is $V \sim r^3/\ell_W$ for a one-dimensional meeting graph. An ellipsoid has the same volume exponent as a ball. The exponent 3 and the ratio ℓ_W are independent readings of (5).

Proposition 12. *$\ell_W = 1$ forces $\lambda = 1/4$. The stub fraction $1 - \exp(-2\lambda)$ is then $1 - e^{-1/2} \approx 0.393$, which the weave preregistration labels overconnected. Every woven ladder value $\lambda \leq 0.08$ has $\ell_W \geq 1.77$.*

The same ℓ_W is the crossover of $d_{\text{eff}}(r)$ and, if stitches are later removed, the transverse factor in any scale read off (5). A scalar $a \propto 1/\rho_S$ built from this metric tracks $a_\perp$, not an isotropic scale factor. The in-plane factor stays 1 while the sheet lattice is held fixed. The ledger average $M_{ab} = \langle n_a n_b \rangle$ needs normals in an ambient three-space. The chain construction does not have those normals. An isotropic plane process would be a different ensemble, and Proposition 11 would not apply to it.

On a finite ring the third direction is compact, circumference $S\ell_W$. For $S = \infty$ the window is the infrared and the axis ratio remains. Spectral dimension of the homogenized walk is 3 only inside the time window corresponding to (4), and returns to 2 after the ring is explored. That return is the compactification, not a second mechanism.

4 One quartic ratio

At $J = 1$,

$$\varepsilon_{J_2}(q) = -8 + 2|q|^2 - \frac{1}{6}(q_x^4 + q_y^4) + O(q^6), \quad (6)$$

$$\varepsilon_{J_3}(q) = -12 + 2|q|^2 - \frac{1}{6} \sum_i q_i^4 + O(q^6). \quad (7)$$

Both are sums of functions of one momentum component, so every coefficient of $q_i^2 q_j^2$ vanishes. The Hessian at the band bottom is $4I$.

Theorem 13. *Write the quartic part as $\beta \sum_i q_i^4 + \gamma \sum_{i < j} q_i^2 q_j^2$. A quartic function of $|q|^2$, and the degree-4 term in $\sqrt{m^2 c^4 + c^2 p^2}$, both require $\gamma/\beta = 2$. For (6), $\beta = -1/6$ and $\gamma = 0$. No choice of m , c , or J removes the mismatch. The first rotationally anisotropic correction and the first departure from the relativistic series sit at the same order.*

The long-wave ray direction can still be radial. On J_2 , $v_i = 4q_i - (2/3)q_i^3 + \dots$, so v is parallel to q on the axes and the diagonals, and the deflection elsewhere is $O(q^2)$. The Manhattan cone is a different object: for $\varepsilon = -4 \sum_{i=1}^d \cos q_i$,

$$\frac{v_E^{\max}}{v_M} = \frac{1}{\sqrt{d}},$$

equal to $1/\sqrt{2}$ on J_2 and $1/\sqrt{3}$ on J_3 . A threshold on the peak and a threshold in the forerunner read these two speeds. The ratio does not require a continuum fit.

A chain weave has Hessian $\text{diag}(4, 4, t_\perp)$ at quadratic order. Rescaling the sheet-index coordinate matches this to $4I_3$ and does not match the quartic, nor the flat band. The J_3 zero eigenvalue is structural at every crystal momentum. A stitch with $\|P_- H P_+\| \neq 0$ lifts the corresponding sheet zero. Nullity against stitch density is the spectral distinction; a quadratic packet, after the rescaling, does not see it.

Dimension universality in the earned sense is universality of the quadratic form: SCALE-0 has $d_H \rightarrow 2$ with J_2 and the square control indistinguishable. Fringes at a few cells see $O(q^2)$ corrections and the plaquette. Two graphs with the same Hessian and different γ/β agree on d_H and need not agree on interference. That is why a spectral feature vector can fail to classify F while still classifying the volume exponent.

5 Band edge, dimension, and a gap

Near a band edge whose Hessian is a multiple of the identity,

$$\rho(E) \propto \begin{cases} (E - E_0)^{-1/2} & d = 1, \\ \text{const} & d = 2, \\ (E - E_0)^{d/2-1} & d \geq 2. \end{cases} \quad (8)$$

The J_2 dispersive edge is the $d = 2$ case. With $m^* = 1/4$ and $J = 1$, $\rho \rightarrow m^*/(2\pi) = 1/(8\pi)$ per quotient cell. The J_3 edge is the $d = 3$ case, vanishing as $\sqrt{E - E_0}$.

Fermi's golden-rule width $\Gamma = 2\pi|V|^2\rho(E)$ therefore stays finite as $E \downarrow E_0$ on J_2 and can be parametrically small on J_3 . Dimension 3 is the lowest dimension in (8) with a vanishing edge density, and it is also the weave value $d_{\text{obs}} = d_{\text{sheet}} + d_M$ for a 2-dimensional sheet and a 1-dimensional meeting complex. The density of states does not forbid $d \geq 4$. The weave count does, and only as a minimality reading: P0' imports Maxwell dimension 2 when it sets $\langle z \rangle = 4$, and the graph alone does not force that integer.

Proposition 14. *On infinite J_2 the essential spectrum of $H = -A$ is $[-8, 8]$ together with the flat band at 0. A finite edge perturbation does not move the essential spectrum. If a vertex subset induces a clique K_n with $n \geq 10$, the normalised indicator of that subset has Rayleigh quotient $-(n-1)$ for H , so*

$$E_0 \leq -(n-1) \leq -9, \quad m = -8 - E_0 \geq n-9.$$

For $n \leq 9$ this trial vector does not clear the edge. A triangle scores -2 , inside the band. Modes inside the essential spectrum hybridize with (8). You could regard a motif with no eigenvalue below the edge as a resonance of the earned unitary, and a motif with such an eigenvalue as a spectral bound state. The flat band on pure J_2 is a protected kernel, extensive in the sheet-antisymmetric sector, and it does not propagate in the quotient.

The event account $R = A_{\text{coef}} + |d|^2/2 + \Re(\bar{d}W)$ telescopes along a sequence of merges to the change in known energy. Between events E_ψ is conserved. Q-DYN-0b treats R as event-local, not as a term stored in Q . The spectral gap of Proposition 14 depends only on the final graph and the final eigenmode. Two histories that reach the same (G, ψ) share that gap and need not share $\sum R$.

At fixed ψ , an edge-set change shifts E_ψ only through $-2\sum(\Delta A_e)B_e$. There is no extra term in $H = -A$ coupling to d_{eff} . Deleting stitches on which $B_e = 0$ changes ball growth and does not change E_ψ . A mass read from the eigenvalue and a dimension read from the ball are independent at that locus.

Isolated K_n has spectrum $\{-(n-1)\}$ once and $\{+1\}$ with multiplicity $n-1$. Only the ground level can sit below -8 , and only for $n \geq 10$. The level $+1$ lies in $[-8, 8]$.

6 Codimension along a unitary orbit

Between events, $i\partial_t\psi = H(G)\psi$ with G and Q fixed, and $\partial H/\partial Q = 0$. VACEXC0 gives the same carrier evolution on every JOINT background. Arrival times read from G , or from the linear packet, are constant on each inter-event interval. Any scale factor built from those times has $\dot{a} = 0$ there. A frozen texture in the flat band does not move H . Orientations that are functions of G are likewise constant. A deterministic drift of M_{ab} would be an update rule for G , which the present equations do not contain. REWIRE0 leaves degree-preserving re-pairings tied, so there is also no bond drift that would rotate a frame.

Unweighted edges have combinatorial length 1. A non-integer dilation is not an automorphism of a fixed simple graph.

Lemma 15. *Across a sum-map split let μ be the difference between the split-side sum $p + q$ and the merged-side amplitude at the same vertex. Then $\mu(t_0) = 0$ for every fibre point, and*

$$\dot{\mu}(t_0) = i \left(s + \sum_{v \in C} \psi_v \right), \quad (9)$$

which does not depend on $d = \psi_i - \psi_j$. The d -dependent piece of $\ddot{\mu}(t_0)$ is $-\frac{1}{2}d(|X_i| - |X_j|)$.

Proof. From $i\partial_t\psi = -A\psi$ one has $\partial_t\psi = iA\psi$. On the split graph,

$$\partial_t(p + q) = i \left(s + \sum_{X_i} \psi + \sum_{X_j} \psi + 2 \sum_C \psi \right).$$

On the merged graph the exclusive and common neighbours each appear once. Subtracting cancels the exclusive sums and leaves (9). Differentiating the exclusive sums brings down $|X_i|p$ and $|X_j|q$, and $p - q = d$ produces the second-derivative coefficient. $\square$

If $s + \sum_C \psi \neq 0$, no choice of d restores C^1 continuity of the summed amplitude. A uniform nonzero state fails (9) on every cover. A STORE replay meets every order because the recovered state is the predecessor, not because d was tuned. An asymmetric cover that has already passed (9) fixes at most one d at the next order. Higher orders then overdetermine it.

Equation (9) is two real constraints. A real condition $B_{ij}(t) = B_*$ is one. Along the unitary orbit, $B_{ij}(t)$ is almost periodic, with frequencies equal to eigenvalue differences of H . For a two-mode parent $B(t) = B_0 + C \cos(\omega t + \varphi)$, crossings of a level inside the range exist and sit at an arccosine comb. A one-dimensional orbit misses a codimension-2 submanifold for generic initial data, and meets a codimension-1 level set in isolated roots. The ratios inside B_* are unfixed (the conditional conservation identity leaves two ratios free). The family of roots, or of Rice rates

$$\lambda(B_*) = \frac{1}{2\pi} \sqrt{\frac{m_2}{m_0}} \exp\left(-\frac{(B_* - \mu)^2}{2m_0}\right)$$

for a stationary Gaussian approximation, is determined by the spectral measure of $B_{ij}(t)$ once the level is given. A two-mode parent is the comb, not a constant hazard. The exact codimension-2 condition has rate zero.

Radiative decay of an in-band resonance is a third clock, with width controlled by (8), and it does not read Q . You could separate a parent with an empty store and an eigenvalue inside the band from a parent that is spectrally bound and carries a stored ξ . The first clock is the unitary width. The second, if it splits, replays ξ and the daughters are the STORE preimage. The marginal of P_- on a Haar pair is Theorem 5, with no free parameter. A departure from that law, from the arcsine law, and from a point mass at $1/2$ would be a different distribution. Agreement with one of the three is the corresponding ensemble.

7 Expansion identities that assume a metric

If a three-dimensional metric is already expanding and the sheets are comoving, area per physical volume scales as a^{-1} , pairwise intersection length as a^{-2} , and triple-point density as a^{-3} . The

relations $a \propto \rho_S^{-1}$ and $a \propto \rho_I^{-1/3}$ are those two dilutions. They do not determine $\ddot{a}$. Given an independent $\ell(t) \propto t^\alpha$, a single flat-universe fluid would have

$$q = \frac{1 - \alpha}{\alpha}, \quad w = \frac{2 - 3\alpha}{3\alpha}.$$

No earned equation produces α . In particular $w = -1$ is not implied by $\dot{\rho}_S < 0$.

On the chain, the derived spacing is $\ell_W \propto \lambda^{-1/2}$. The exponent $-1/3$ belongs to triple points in the three-volume, not to the stitch intensity λ . While λ is fixed, Section 6 gives $\dot{a} = 0$.

For a pure power $V = A(r/\ell)^d$, $d_{\text{eff}} = d$ is independent of ℓ . On the chain, d_{eff} is a function of r/ℓ_W only: 2, then 3, then 2 again past $S\ell_W$. Walking in r on a fixed graph traces that curve. A time-dependent exponent at fixed probe radius needs $\ell_W(t)$ or $S(t)$. The local spectral gap of a tense motif stays fixed in hopping units if local edges are held fixed while ℓ_W grows. The probe remains inside the three-dimensional window only while $\ell_W \ll r_{\text{probe}} \ll S\ell_W$.

Static response of a drive detuned by $1/2$ below the J_2 edge is the massive symbol $1/2 + 2|q|^2$, Helmholtz constant $\kappa = 1/2$. The gapless Laplace response sits at the band edge, which the electromagnetic falsification labels as excluded tuning. On a three-dimensional quotient that excluded Green function would be $1/r$. J_2 is 8-regular against the Maxwell value 4, so excess coordination is already present. Curvature of (5) is the curvature of an anisotropic background whose axis ratio is ℓ_W , not a linearization about δ_{ab} , and only where ℓ_W varies in space. A uniform ℓ_W has none of that curvature. Between events a frozen Q does not source it, because $\partial H/\partial Q = 0$.

8 Plaquettes as an order parameter

A tree has exponential ball growth and no integer plateau of d_{eff} . An expander has diameter $\sim \log N$. The earned geometric graphs sit between them, with a density of 4-cycles of order one per vertex. You could take the fraction $f_\square$ of edges that lie in a 4-cycle as a coordinate on that interval: 0 on a tree, order one on the square and on J_2 . Across a percolation of those cycles at fixed degree, a logarithmic derivative that was large on the tree side falls toward 2. A later weave window at 3 is Section 3, and it is intermediate when S is finite. A rise of d_{eff} from 1 toward 3 is a different path through the same $(f_\square, d_{\text{eff}})$ plane.

Part II

Local derivations

9 Isolated information of one stored pair

The open row asks for an entropy built from exterior-blind store entries. The pair data (s, d) already determine a single real weight. Lemma 1 gives

$$P_- = \frac{1}{2} - \frac{B_{ij}}{q_{ij}}$$

whenever $q_{ij} > 0$, and Corollary 2 identifies the maximum of $h_2(P_-)$ with a norm-neutral bond. Remark 3 keeps h_2 optional: the ledger has not introduced $-\sum p \log p$.

Theorem 5, Proposition 6 and Proposition 7 are three exact means on one pair: $1/(2 \ln 2)$, $2 - 1/\ln 2$, and 0. Proposition 8 is the scale-separated limit, in which $h_2 \rightarrow 1$ because the amplitudes

differ, not because a phase was selected. The same P_- is the daughter weight in the split fibre (Section 11) and the summand in any cut sum that one might compare with the I1 bit (Section 1). The continuous dimension of d is Section 2 and is not this sum.

10 Dimension of the exterior-blind store

Lemma 9 is the local count: one generic merge hides a complex line, real dimension 2, dropping to 1 only in the all-zero merged state. Proposition 10 is the global count under a boundary-sized imprint:

$$D_Q \geq 2(N_Q - b),$$

volume-leading when $N_Q = n_R - 1$. If Q does not enter H , the Jacobian vanishes and the kernel is the full $2(n_R - 1)$, up to all-zero exceptions.

You could compare this with $\sum h_2(P_-)$ from Section 1. The comparison is an inequality of different quantities. An area law for one does not force an area law for the other. The factor 2 is $\dim_{\mathbb{R}} \mathbb{C}$. I1a's one bit per saturated leg is a third normalization, available only after an edge set of cardinality k and a mean $\bar{h}$ have been chosen.

11 What of $\xi = (c, d)$ is already fixed by B

A daughter that is not a stored replay has fibre coordinate $d \in \mathbb{C}$ and a discrete cover. The combination that enters Remark 3 is only the real number $P_- = 1/2 - B/q$. The phase of d at $B = 0$ is the current and is invisible to P_- . So the fibre splits into the cover, the bond, and the current.

A stored entry fixes all three (STORE). A Haar pair, with nothing stored, has P_- uniform on $[0, 1]$ by Theorem 5, with no free parameter in that marginal. A scale-separated pair has $P_- \rightarrow 1/2$ by Proposition 8, and the phase remains free unless the current or the jet fixes it. The 520-parameter fibre measure is a law on the whole fibre compatible with every earned constraint. The marginal of P_- under a Haar draw is narrower than that, and it is already computed. Lemma 15 constrains which covers can meet C^1 continuity at all.

12 Daughters from a consumed entry

Given $\xi = (c, d)$ and the merged state, the sum-map preimage is unique up to the swap $d \mapsto -d$. Between events the entry does not change, so the entry present at a split is the entry written at the merge. Several entries leave the discrete choice of which entry. Lemma 15 admits at most those entries whose cover satisfies $s + \sum_C \psi = 0$; an asymmetric survivor fixes at most one d . An empty store returns the situation of Section 11. The three parameter-free marginals on P_- are the Haar law, the arcsine law, and a point mass at $1/2$.

13 Unitary width and a replay of ξ

An eigenvalue inside the essential spectrum couples to the continuum (8). The golden-rule width is $\Gamma = 2\pi|V|^2\rho(E)$. On the J_2 edge, ρ tends to a positive constant, so the width does not shut off as the resonance approaches -8 . On the J_3 edge, ρ vanishes as a square root. This evolution is e^{-itH} . It does not read Q .

A sum-map split that consumes a stored ξ returns the STORE preimage. A split that creates ξ is Section 11, and Lemma 15 removes every cover that fails (9), independently of d . A motif with no eigenvalue below the edge (Proposition 14 and the remark after it) has the unitary clock available and has no spectral gap to protect a later structural split.

14 Event account against spectral gap

Each merge has the event-local value

$$R = A_{\text{coef}} + \frac{|d|^2}{2} + \Re(\bar{d}W) = -(\Delta E_\psi + \Delta E_G),$$

and the reverse split carries $-R$. Along a sequence, $\sum_i R_i$ equals the change in known energy, because E_ψ is conserved between events. Radiation that leaves the region is part of that final known energy. The gap $m = -8 - E_0$ of Proposition 14 depends only on the final graph and the final eigenmode. Two sequences that arrive at the same (G, ψ) share m and need not share $\sum R$. Norm change of the same event is $2B$, which is Section 1 and is not a firing condition.

15 First derivative across a sum-map split

Lemma 15 is the statement. The first derivative of the summed amplitude jumps by $i(s + \sum_C \psi)$ for every value of the fibre coordinate d . The cover enters only through the common-neighbour set C . The second derivative sees d with coefficient $-\frac{1}{2}(|X_i| - |X_j|)$. A uniform nonzero vacuum fails the first-order condition on every cover. A stored replay satisfies every order because it is the predecessor. You could treat (9) as a necessary condition on any jet match that is meant to be C^1 for the summed amplitude. It is two real constraints, hence the codimension discussed in Section 6, and it is the same s that appears in P_- .

16 Roots of a bond threshold

A condition $B_{ij}(t) = B_*$ is one real constraint on an almost-periodic function whose frequencies are eigenvalue differences of H . For $B(t) = B_0 + C \cos(\omega t + \varphi)$ the crossings, when they exist, are

$$t = \frac{1}{\omega} \left(\pm \arccos \frac{B_* - B_0}{C} - \varphi \right) + \frac{2\pi n}{\omega}.$$

Equation (9) is two real constraints. A generic one-dimensional orbit misses it, which is compatible with a unitary flow that does not fire and with perturbations that stay small. Several edges meeting a codimension-1 level in the same window leave a tie of the same kind as a degenerate re-pairing (Section 19). The level B_* still contains the two unfixed ratios of the conditional conservation identity. The crossing times of every candidate level are functions of the single process $B_{ij}(t)$.

17 Three clocks already fixed by H

Section 6 separates three waiting laws that do not require a new measure. The radiative width is $\Gamma = 2\pi|V|^2\rho(E)$ with ρ from (8). The bond threshold is the arccosine comb of Section 16, which on a few-mode parent is a train of steps. The exact codimension-2 jet condition is missed, so its structural rate is zero. A constant hazard appears in the further approximation that $B(t) - B_*$ is

a mixing Gaussian process and events are upcrossings. That rate is the Rice formula in Section 18, with moments taken from the spectral measure of the same edge process. The parent that is spectrally bound and carries a stored ξ is the case in which the structural clock can be the replay of Section 12 rather than the radiative width.

18 Level dependence of a Rice rate

For a stationary process $X(t) = B_{ij}(t) - \mu$ with variance m_0 and derivative variance m_2 , the mean upcrossing rate of a level B_* is

$$\lambda(B_*) = \frac{1}{2\pi} \sqrt{\frac{m_2}{m_0}} \exp\left(-\frac{(B_* - \mu)^2}{2m_0}\right).$$

Both moments are integrals against the spectral measure of $B_{ij}(t)$. The family $\lambda(B_*)$, not a single rate, is the content of the formula. A two-mode parent replaces the exponential by a sum of deltas at the arccosine roots. A single Haar vector determines the law of the snapshot B/q (Theorem 5) and does not determine m_2 . The exact codimension-2 condition of Lemma 15 has $\lambda = 0$. The unfixed ratios inside B_* are the same debt as in the conditional conservation identity. A rate independent of the level would lie outside this formula.

19 A tie with no bond drift

A degree-preserving re-pairing changes the local isomorphism type at

$$\Delta E = -2(B_{\text{add}} - B_{\text{remove}}).$$

When the two completions of a 4-cycle have the same ΔE , the bond does not select. REWIRE0 finds the same tie, or an absence, under the frozen exact principles, persistently in the linear size. A branching ratio would be a measure on that tie. No such measure is forced by the earned constraints. You could still compare the two rewired graphs by the single number ΔE , on a substrate other than J_2 . A splitting that stays finite as the linear size grows would be a scale in the same units as B . A splitting that closes leaves the tie flat. That number is the same B as in Section 1.

20 An eigenvalue below the essential spectrum

Proposition 14 is the sufficient condition: an embedded K_n , $n \geq 10$, forces $m = -8 - E_0 \geq n - 9$ on J_2 , in hopping units at $J = 1$. The same Rayleigh quotient on J_3 is compared with the edge -12 . For $n \leq 9$ the all-ones trial vector does not clear -8 . In-band eigenvalues hybridize with the density (8), which is the width in Section 13. The flat band at 0 is a kernel of the sheet-antisymmetric sector. A packet supported there does not propagate in the quotient. A quotient-moving mode sits in the dispersive band, and it remains an eigenvalue outside that band only if some trial vector pushes it below the edge.

21 Gap in hopping units

The normalised uniform mode of 8-regular J_2 sits at eigenvalue -8 , independent of the number of vertices. Define $m = -8 - E_0$ when $E_0 < -8$ and $m = 0$ otherwise. Proposition 14 gives $m \geq n - 9$ for an embedded clique of order $n \geq 10$. The trial vector uses only the edges inside its

support. Extra edges from that support to the bath do not raise the Rayleigh quotient. Removing edges lowers the all-ones quotient toward the interior of the band, so the same trial vector does not produce a gap from under-coordination. This m is an eigenvalue difference, not an extensive energy: $E_\psi = \langle \psi | H | \psi \rangle$ for a normalised state remains $O(1)$. A conversion into a rest energy in the units of I1 is a further identification. The dimensionless difference $E_0 + 8$ does not need it. The same gap is what Section 14 distinguishes from $\sum R$.

22 Ball growth at fixed field

For two graphs on the same vertex set and one fixed ψ ,

$$E_\psi(G') - E_\psi(G) = -2 \sum_{e \in E' \setminus E} B_e + 2 \sum_{e \in E \setminus E'} B_e.$$

No term in $H = -A$ depends on d_{eff} except through the edge set. If every deleted stitch has $B_e = 0$, the field energy is unchanged and the ball growth may still change. At that locus a dimension and the energy of Section 21 separate. The bond $B_e = 0$ is the same point as $P_- = 1/2$ in Lemma 1. A frustration energy proportional to $|d_{\text{local}} - 3|$ would be a reparameterization of the bond sum, and only where the deleted edges happen to carry B_e correlated with that difference. The identity above does not force the correlation.

23 Eigenvalues of a finite motif

A motif on n vertices has at most n eigenvalues of $H = -A$. Those below the essential-spectrum edge are stable against the continuum (8). Those inside the band are resonances whose widths track $\rho(E)$. Isolated K_n has one level $-(n-1)$ and an $(n-1)$ -fold level $+1$. Only the ground level can lie below -8 , and only for $n \geq 10$. The level $+1$ lies in $[-8, 8]$. Coupling to the lattice can split that multiplet into resonances. It pushes a vector below the edge only if some trial vector says so; the all-ones direction is the one that scores $-(n-1)$. Commutant of the motif automorphism group with H gives the selection rules among those eigenvalues. This spectrum is the internal structure whose ground gap is Section 21.

24 Vanishing edge density against a weave count

The density (8) vanishes at a nondegenerate band edge if and only if $d \geq 3$. The weave reading $d_{\text{obs}} = 2 + d_M$ equals 3 when the meeting complex is one-dimensional. These are independent. The first forbids a finite golden-rule width at threshold (Section 13). The second is minimality among polynomial meeting graphs, and it uses the imported Maxwell dimension 2. Neither statement is implied by the other. A meeting dimension $d_M \geq 2$ gives $d_{\text{obs}} \geq 4$, which still has a vanishing edge density. The clique gap of Proposition 14 can be formed in any of these dimensions; the width of whatever remains inside the band cannot.

25 Quartic symbol of a separable dispersion

Theorem 13 is the obstruction. Axis-aligned nearest-neighbour $H = -A$ has $\gamma/\beta = 0$ at the band bottom, while a rotationally invariant quartic and the relativistic series both have $\gamma/\beta = 2$. The quadratic piece can still be matched to $p^2/(2m)$ with $m = 1/(4J)$. The group of that quadratic

symbol is the Schrödinger group. The degree-4 term breaks rotations, so the failure is not only a failure of boosts. On J_2 the ray deflection away from the axes and the diagonals is $O(q^2)$, which is the same order. A substrate whose hoppings are not axis-aligned can produce $\gamma \neq 0$. Those graphs are the ones that leave the square class in the interference gates. The ratio is also what survives a quadratic rescaling between J_3 and a weave (Section 27).

26 Two speeds from one separable symbol

For $\varepsilon = -4 \sum_{i=1}^d \cos q_i$ the maxima of group velocity satisfy

$$\frac{v_E^{\max}}{v_M} = \frac{1}{\sqrt{d}}.$$

The Manhattan value is the causal bound of a sum of axis hops (8 on J_2 , 12 on J_3). Near $q = 0$, $v = 4q + O(q^3)$ is parallel to q up to the deflection in Theorem 13. A localized source, superposed, fills a Euclidean ball at the peak and a diamond at the support of the forerunner. The ratio $1/\sqrt{d}$ is the relation between those two speeds. It is the same separability that forces $\gamma = 0$ in Section 4.

27 Quadratic match against a lifted zero

The J_3 Hessian at the band bottom is $4I_3$. A chain of J_2 sheets with transverse scale $t_\perp$ has $\text{Hess} = \text{diag}(4, 4, t_\perp)$. The coordinate change $z' = z\sqrt{t_\perp/4}$ identifies the quadratic forms and leaves the quartic ratio of Theorem 13 equal to 0 in the plane, with a different transverse quartic.

The J_3 flat band is the zero eigenvalue of $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ at every crystal momentum. A stitch with a nonzero matrix element between the sheet-symmetric and sheet-antisymmetric vectors lifts the corresponding zero. At small stitch amplitude the splitting is linear for a coherent stack and of the order of the transverse bandwidth for a disordered weave. A disjoint union of S sheets starts with SL^2 flat zeros. After the stitches those zeros are no longer the J_3 count L^3 plus touchings. Section 3 is the metric form of the same $t_\perp$, read as ℓ_W rather than as a hopping.

28 Quadratic form against a fringe

SCALE-0 identifies the volume exponent of J_2 with that of the square, to machine precision, out to large L . Both have Hessian $4I$ at the band bottom. Any long-wave generator that is a multiple of the Laplacian shares that exponent. The relative size of the first correction is $O(q^2) = O((a/r)^2)$. At distances of a few cells the quartic ratio γ/β and the plaquette holonomy are order one. Theorem 13 says that ratio is zero on the separable lattices and 2 on a rotationally invariant symbol. Two graphs can therefore share d_H and differ at the slit. That split is consistent with a feature vector that classifies the volume exponent and does not classify interference.

29 Transport metric from the Hessian

The long-wave packet speed is the inverse of the Hessian of $\varepsilon(q)$ at the band bottom. On J_2 and on J_3 that Hessian is $4JI$, so the quadratic transport metric is conformal to the lattice metric, with speed fixed by J . A stitch density changes the transverse entry (Section 27) and therefore changes that metric. A flat-band texture does not enter H and does not move arrival times. The

quartic part of ε is invisible to this metric and visible to the angular deficit of Theorem 13. A time-dependent transport metric requires a time-dependent H , hence a time-dependent graph. Between events the metric extracted this way is constant (Section 6).

30 Volume window of a chain of sheets

Sheets of intrinsic area growth r^2 , meeting graph of dimension d_M , and crossover ℓ_W give

$$V(r) \sim \begin{cases} r^2 & r \ll \ell_W, \\ (r/\ell_W)^{d_M} r^2 & \ell_W \ll r \ll R_\perp, \\ S r^2 & r \gg R_\perp, \end{cases} \quad (10)$$

with $R_\perp = S\ell_W$ when $d_M = 1$ and the meeting graph is a ring of S sheets. In that case the middle regime is $V \sim r^3/\ell_W$.

On an infinite line of sheets the middle regime is the infrared: $V \sim r^3/\ell_W$ for every $r \gg \ell_W$. The return to thick two-dimensional growth is the finite- S compactification, circumference $S\ell_W$. A finite ring that shows the exponent 3 only inside a window has exhibited (10). Absence of the re-entrant tail on that ring means the probe has not reached $r \sim S\ell_W$.

A dense meeting graph, $d_M = \infty$, has no middle regime: $V \sim S r^2$ as soon as the stitches are felt. A 3-regular random meeting graph is locally a tree. Its volume is exponential in graph distance and has no stable power 3.

The exponent 3 and the axis ratio ℓ_W are independent readings of (5). An ellipsoid has the same volume exponent as a ball. Section 27 is the spectral form of the same distinction: a quadratic packet, after a transverse rescaling, does not see the lifted sheet zero.

31 One length for the spectral window and the ball

The stitch spacing (4) is the only transverse scale in the chain construction. A random walk with in-plane diffusion constant $D_\parallel$ and transverse constant $D_\perp$ in the sheet-index coordinate has return probability

$$Z(t) = (4\pi D_\parallel t)^{-1} (4\pi D_\perp t)^{-1/2} \quad (11)$$

throughout $\ell_W^2/D_\parallel \ll t \ll (S\ell_W)^2/D_\perp$. The spectral dimension $-2 d \ln Z / d \ln t$ equals 3 inside that window, equals 2 at shorter time, and returns to 2 after the ring is explored. The constant $D_\perp$ absorbs one factor of ℓ_W^2 if the walk is written in a coordinate where each stitch is a unit step.

Graph distance is a different functional. A transverse coordinate step of size 1 costs graph length ℓ_W . The radius where $d_{\text{eff}}(r) = d \ln V / d \ln r$ crosses 2.5 therefore tracks the same $\ell_W(\lambda)$ that sets the window of (11), and the ratio of the principal radii of the ball tracks ℓ_W as well (Proposition 11). You could regard a ball whose crossing radius follows ℓ_W while its axis ratio does not as evidence that the volume exponent has a source other than this homogenization.

32 In-plane disorder leaves the transverse eigenvalue

Proposition 11 is the local statement. Random shifts and a transpose act on (x, y) . The coefficient ℓ_W^2 of dz^2 is invariant. The sheet Hessian is already $4I$, so those moves have no quadratic in-plane anisotropy to remove, and they do not equalize the transverse eigenvalue with the in-plane ones.

Proposition 12 places the isotropic point $\ell_W = 1$ at $\lambda = 1/4$, stub fraction $1 - e^{-1/2} \approx 0.393$, which the weave preregistration calls overconnected. Every woven value $\lambda \leq 0.08$ keeps $\ell_W \geq 1.77$. The volume exponent can equal 3 on that range (Section 30) while the axis ratio stays at least 1.77.

The average $M_{ab} = \langle n_a n_b \rangle$ needs normals in an ambient three-space. The chain construction does not assign them. You could read an isotropic infrared three-space, built from sheets, as a meeting complex whose transverse and in-plane costs match. On (5) that match is $\ell_W = 1$, which removes the scale separation the crossover is built on.

33 A noncompact meeting direction

A ring of finitely many sheets is three-dimensional only inside the window of (10) and is a thick two-dimensional stack in the infrared. A stable infrared exponent 3, built from locally two-dimensional sheets, uses one of the following.

An infinite line, $S = \infty$, with ℓ_W finite. Then $V \sim r^3/\ell_W$ for every $r \gg \ell_W$. The third direction is noncompact, and the metric remains anisotropic by ℓ_W (Section 32).

A meeting complex that is itself a noncompact three-dimensional foam, so that no axis stays at a parametrically different cost. That complex is not the chain construction, and Proposition 11 does not apply to it.

A finite stack, however large, has compactification radius $S\ell_W$. The extrapolation of the chain past that radius is the infinite-line formula, which still carries the axis ratio. Section 34 records what that ratio does to a scale factor.

34 A scale factor that the carrier does not generate

Between structural events the carrier is $i\partial_t\psi = H(G)\psi$ with $H = -A$ and Q frozen. Graph distance, resistance distance, and linear arrival time are properties of G . They are constant on each inter-event interval, so any scale factor read from them has $\dot{a} = 0$ there (Section 6). A flat-band texture does not enter H and does not evade this. Unweighted edges have combinatorial length 1. A non-integer dilation is not an automorphism of a fixed simple graph.

If a three-dimensional metric is already expanding, the dilutions $a \propto \rho_S^{-1}$ and $a \propto \rho_I^{-1/3}$ describe that expansion. They do not determine $\ddot{a}$. An independent $\ell(t) \propto t^\alpha$ would give

$$q = \frac{1 - \alpha}{\alpha}, \quad w = \frac{2 - 3\alpha}{3\alpha}$$

for a single flat-universe fluid. No earned equation produces α . In particular $w = -1$ is not implied by $\dot{\rho}_S < 0$.

On the chain, growth of ℓ_W at fixed sheet lattice is

$$a_{\parallel} = 1, \quad a_{\perp} \propto \ell_W.$$

The Hubble anisotropy is then of order one throughout the woven window $\ell_W \geq 1.77$. The kinematic point $\ell_W \rightarrow 1$ erases the two-dimensional window and lands on the overconnected value of Proposition 12. You could regard that point as a three-dimensional substrate, in the sense of Section 27, rather than as an equation of state.

35 Sheet density tracks one axis

Take the three-volume to be the anisotropic ball of (5), and take ρ_S to be sheet area per that volume. With the in-plane cell area held fixed, ρ_S scales as $1/\ell_W$. The identification $a \propto \rho_S^{-1}$ then means $a \propto \ell_W$, which is $a_\perp$. The in-plane factor stays 1.

Under that identification $H = -\dot{\rho}_S/\rho_S$ is the transverse Hubble rate. It vanishes while λ is fixed (Section 34). A discrete deletion of stitches changes λ in steps. After n such steps the transverse factor is $a_\perp(n) \propto \lambda_n^{-1/2}$, with $a_\parallel$ unchanged. A scalar fitted to the mean of the three principal radii moves when two axes do not. That scalar is a different object from the metric (5).

36 Which intersection fixes the exponent

Hold a comoving population of surfaces inside an already expanding isotropic volume, as a kinematic comparison. Area density scales as a^{-1} . Length density of pairwise intersection lines scales as a^{-2} . Number density of triple points scales as a^{-3} . The relation $a \propto \rho_I^{-1/3}$ is the triple-point count. Line density would give $a \propto \rho_L^{-1/2}$. Sheet area would give $a \propto \rho_S^{-1}$.

The stitch intensity λ is a number of stitches per node. That is a point process on the sheet, not a triple point in the three-volume. On the chain the derived ruler is $\ell_W \propto \lambda^{-1/2}$ (4), so the transverse scale tracks the exponent $-1/2$. Replacing $-1/2$ by $-1/3$ changes $a_\perp(\lambda)$ by a power. You could keep the Poisson exponent unless a different intersection is the one being counted, and name that intersection when the exponent changes. Section 7 is the same separation, written without a preferred density.

37 A pure power against a window

For $V(r) = A(r/\ell)^d$ with A , d , and ℓ independent of r ,

$$d_{\text{eff}} = d, \quad \frac{\partial d_{\text{eff}}}{\partial \ell} = 0.$$

Changing ℓ at fixed d is a pure rescaling. On a graph whose every edge has length 1, arrival times in hopping units scale with the distances. The ratio of a graph clock to a graph ruler stays zero.

The ratio becomes visible when one clock does not scale. The gap m of Section 21 is set by local coordination. If those local edges are held fixed while ℓ_W grows, m stays constant in hopping units and ℓ_W grows in the same units. The window exponent 3 is unchanged for a probe that remains inside

$$\ell_W \ll r_{\text{probe}} \ll S\ell_W.$$

A probe at a fixed number of cells eventually leaves that window as ℓ_W grows, and the local reading returns to 2. The inequality is the condition under which d_{eff} and a growing transverse scale are compatible. On the woven ladder $\ell_W \geq 1.77$, so the window, measured in units of ℓ_W , is the finite range up to S .

38 A static curve in r/ℓ_W

On a pure power the logarithmic derivative is the constant d , independent of ℓ . On the chain it is a function of the dimensionless ratio r/ℓ_W :

$$d_{\text{eff}}(r) \rightarrow \begin{cases} 2 & r/\ell_W \rightarrow 0, \\ 3 & 1 \ll r/\ell_W \ll S, \\ 2 & r/\ell_W \gg S. \end{cases} \quad (12)$$

An observer at fixed graph distance sees a fixed exponent. An observer who changes the probe scale, on a fixed graph, walks along (12). That is a scale dependence. A time dependence at fixed probe radius needs $\ell_W(t)$ or $S(t)$, hence a changing graph (Section 34).

Until the graph changes, the small- r and large- r ends of (12) are two ends of one static curve. You could collapse d_{eff} against $\log(r/\ell_W)$ for several λ . A collapse onto one curve means the single length (4) controls the crossover. A failure to collapse means that formula does not. Section 31 is the same length read from the return probability rather than from the ball.

39 No drift of an orientation matrix

The chain weave has no ambient normals, so $M_{ab} = \langle n_a n_b \rangle$ is not an observable of that ensemble (Section 32). On an ensemble that does assign normals, the orientation of a sheet is a function of G . Between events G is fixed, so M_{ab} is fixed. Degree-preserving re-pairings that could rotate a local frame are the tied moves of REWIRE0: no bond, current, or energy drift selects a rotation (Section 6). The deterministic drift of M_{ab} is zero. A diffusion in orientation space would be a measure on those ties.

Under every earned deterministic rule the eigenvalue gap of M_{ab} is therefore constant in time. Isotropy, when it is present, is a property of the initial ensemble or of the construction. Haar measure on orientations gives $M_{ab} = I/3$ by symmetry. The Hessian of J_3 is $4I$ by the form of ε_{J_3} . Neither is an attractor of $H = -A$. A declining gap of M_{ab} along a trajectory generated by H would mean the angle update was not that Hamiltonian.

40 Cycle density between a tree and an expander

A sparse tree has exponential ball growth, $V(r) \sim (z-1)^r$, until the finite size cuts it off. The spectral dimension of a critical branched polymer is $4/3$. There is no integer plateau of d_{eff} . An Erdős–Rényi graph above the connectivity threshold, or a configuration model with excess degree and large girth, is an expander: the diameter is $\sim \log N$, and a transport estimator that assumes a polynomial ball does not return a stable embedding dimension.

The earned geometric graphs sit between these. J_2 and the square have a density of 4-cycles of order one per vertex. The chain weave adds a thin set of stitches and keeps the sheet cycles. The fraction $f_{\square}$ of edges that lie in a 4-cycle is 0 on a tree, of order one on the square and on J_2 , and small on a dilute random regular graph of large girth.

Across a percolation of those cycles at fixed degree, a logarithmic derivative that was large on the tree side falls toward 2. A later weave window at 3 is Section 3, and it is intermediate when S is finite. You could plot d_{eff} against $f_{\square}$ on one configuration-model family as short cycles are added, with the degree sequence held fixed. The expander side, $f_{\square} \rightarrow 0$ at high degree, does not pass through a plateau at 3 on that path. Section 8 is this coordinate written as the order parameter of the soup.

41 A drop of the reported exponent

Section 40 identifies a geometric phase with a percolation of 4-cycles at fixed degree. On the tree side of that transition the ball grows exponentially, and a local logarithmic derivative $d \ln V / d \ln r$ is large. On the sheet side it equals 2. Formation of geometry, in this order parameter, is a decrease of the reported exponent toward 2, followed by a weave lift toward a window at 3 if stitches percolate in a one-dimensional meeting complex (12).

A path on which d_{eff} rises from 1 toward 3 is a different curve in the $(f_{\square}, d_{\text{eff}})$ plane (Section 8). No earned firing law produces $f_{\square}(t)$. The static family parameterized by cycle density is the phase diagram. A dynamical time is a later parameterization of the same axis, and only if some rule changes $f_{\square}$. The width of the window at 3, when it appears, remains $S\ell_W$.

42 A detuned static symbol

A static drive at frequency w has symbol $\varepsilon(q) - w$. The drive $w = -8.5$ on J_2 sits $1/2$ below the band edge -8 . The infrared symbol is

$$\frac{1}{2} + 2|q|^2, \quad (13)$$

a massive Helmholtz operator with $\kappa = \sqrt{(1/2)/2} = 1/2$. The Green function decays as $e^{-\kappa r}$ times the dimension factor, $r^{-1/2}$ in two dimensions and r^{-1} in three. The operational static channel of the electromagnetic falsification saturates at a few cells. That saturation is this range.

The gapless Laplace response sits at $w \rightarrow -8$. The same falsification labels that tuning excluded. On a three-dimensional quotient the excluded Green function would be $1/r$. A far-zone $1/r^2$ force from the static response of ψ is not a consequence of (13). Enlarging the grid at fixed detuning keeps the range $O(1)$.

J_2 is 8-regular. The Maxwell count imported by P0' is coordination 4. Excess coordination already places J_2 outside the isostatic class in which a stress response can be gapless for a reason other than a tuned drive frequency. The square sheet has $\langle z \rangle = 4$. You could read a long-range structural susceptibility, if one exists, as a property of that rigidity class. Section 5 is the same edge, read as a density of states rather than as a Green function.

43 A store that does not enter H

Q-DYN-0b holds Q constant on every waiting rung. The account R is recomputed when a structural map runs. It is not a term in H . The linear carrier does not see Q . A pure store difference at fixed ψ changes neither B nor the arrival metric. Between events

$$\frac{\partial H}{\partial Q} = 0. \quad (14)$$

There is no secular δG sourced by a frozen Q .

The sentence that disturbances flow into Q and G can describe the bookkeeping at an event: d moves into the store, and R accounts the energy jump (Section 5). It does not describe a force during the waiting interval. Two states with the same (G, ψ) and different Q have identical $\psi(t)$, identical $B(t)$, and identical arrival times until an event is allowed to read Q . A far signal at fixed G would contradict (14). Section 2 counts the dimension of the store that this derivative does not see.

44 Curvature of a varying stitch length

A localized excess or deficit of stitches changes $\ell_W(x)$ and therefore the transverse coefficient in

$$ds^2 = dx^2 + dy^2 + \ell_W(x)^2 dz^2. \quad (15)$$

The curvature of a diagonal metric is a function of $\ell_W(x)$ and its derivatives. The background itself is anisotropic by ℓ_W , and on the woven ladder $\ell_W \geq 1.77$ (Proposition 12). A linearization about δ_{ab} is a different metric.

Perturbations on (15) fall into two classes. At fixed ℓ_W , the in-plane static response is the Helmholtz problem (13), or the excluded Laplace problem at the band edge (Section 42). The range is $O(1)$, or $1/r$ only on that excluded tuning. Gradients of ℓ_W are changes of G . They are absent between events (Section 43).

If ℓ_W is uniform, the curvature associated with its derivatives vanishes and a packet propagates on the constant anisotropic background. A quadrupole aligned with the sheet normal, scaling as the second derivatives of ℓ_W , is present when the stitch field varies in space. An isotropic $1/r$ potential is not a solution of (15). Section 29 is the same transverse coefficient read from the Hessian rather than from the curvature.